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METHODS AND APPARATUS FOR PRESERVATION OF ARTICLES

Abstract

An apparatus includes a container for preservation of articles defining an atmospheric control mechanism that maintains an internal atmospheric composition within the container such that an interior cavity of the container that is devoid of various contaminants including UV radiation, dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the articles to be protected. As such, the container protects and preserves articles by preventing decay and deterioration of the articles due to light, temperature, dirt, mold, humidity, hydrolysis due to moisture in the air, and oxidation with exposure to air.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/836,812, filed Jun. 9, 2022, which claims benefit of U.S. Provisional Application Ser. No. 63/208,771, filed on Jun. 9, 2021, which are herein incorporated by reference in their entirety.

FIELD
[0002] The present disclosure generally relates to containers and/or other similar consumer goods, and in particular, to a system and associated method for an apparatus for preservation of articles stored therein.
BACKGROUND
[0003] Certain articles of value are susceptible to decay and deterioration due to light, temperature, dirt (particulate contamination), mold, humidity, hydrolysis due to moisture in the air, and oxidation with exposure to oxygen. For instance, shoes that have inherent monetary and/or personal value such as Nike Air Force 1s can include plastic materials, color configurations and adhesives that can deteriorate over time due to environmental factors. Other articles of monetary or personal value that are also susceptible to degradation due to environmental factors include collectible cards of various subjects, postage stamps, clothing, electronics, media including tapes, photography film, documents.
[0004] It is with these observations in mind, among others, that various aspects of the present disclosure were conceived and developed.
SUMMARY
[0005] Aspects of the present novel disclosure may take the form of an apparatus including at least one container. The at least one container is configured to maintain an internal atmospheric composition and includes an interior cavity in communication with an opening configured to receive an article. The apparatus further includes an access door that provides access to the interior cavity along the opening of the at least one container, the access door including a sealing mechanism that couples the access door along the opening such that the sealing mechanism prevents permeation of air from an external environment into the interior cavity. The apparatus further includes an inlet valve coupled to the interior cavity of the container and configured to provide one-way fluid flow of a selected gas into the interior cavity to reach and maintain the internal atmospheric composition and preserve the article. The apparatus further includes an outlet valve coupled to the interior cavity that establishes one-way fluid flow communication from the interior cavity to the external environment that further contributes to maintaining the internal atmospheric composition.
[0006] In some examples, the apparatus is configured to: receive an article within the interior cavity of the container through the airtight seal provided collectively by the opening and the access door, assume a sealed configuration wherein the opening is closed or sealed via the access door, establish one-way fluid flow of air from the internal cavity to the external environment by engaging (opening) the outlet valve, receive a selected gas to within the interior cavity through the inlet valve such that undesired components including oil, oxygen, moisture, and the like are purged from the interior cavity through the outlet valve by pressure from the incoming selected gas, and form the internal atmospheric composition whereby the interior cavity is limited or devoid of the undesired components. In some examples, canisters of the selected gas include a predetermined amount of the selected (compressed) gas so as to fill the entire interior cavity, thereby flushing or purging the

interior cavity of the previous air composition. Once the necessary or desired amount of inlet gas is transferred through the inlet valve, the inlet valve and the outlet valve can be shut or closed.
[0007] Aspects of the present novel disclosure may take the form of a method of making an apparatus that preserves an article, comprising the steps of: forming at least one container configured to maintain an internal atmospheric composition, the at least one container including an interior cavity in communication with an opening configured to receive an article; forming an access door along the opening of the at least one container that provides air-tight access to the interior cavity; coupling an inlet valve to the interior cavity that provides one-way fluid flow of a selected gas into the interior cavity; and coupling an outlet valve to the interior cavity that provides one-way fluid flow communication from the interior cavity to the external environment.
[0008] These examples and features, along with many others, are discussed in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a simplified illustration showing a container including an outer container for preservation of articles;
[0010] FIGS. **2**A and **2**B are a series of illustrations showing front, side, and rear views of the container of FIG. **1** in a first “pillow” configuration;
[0011] FIGS. **3**A-**3**C are a series of illustrations showing front, side, and rear views of the container of FIGS. **2**A and **2**B filled with an inert gas and including an article;
[0012] FIGS. **4**A and **4**B are a series of illustrations showing operation of an inlet valve with a canister of the container of FIG. **1**;
[0013] FIGS. **5**A-**5**E are a series of illustrations showing operation of an outlet valve of the container of FIG. **1**;
[0014] FIGS. **6**A-**6**D are a series of illustrations showing the container of FIG. **1** in a second “box” configuration;
[0015] FIG. **7** is an illustration showing an exploded view of the container of FIGS. **6**A-**6**D;
[0016] FIGS. **8**A-**8**C are a series of illustrations showing an access door of the container of FIG. **7** in respective closed and open configurations;
[0017] FIG. **9**A is an illustration showing an inlet valve of the container of FIG. **7**;
[0018] FIG. **9**B is an illustration showing an outlet valve of the container of FIG. **7**;
[0019] FIG. **10** is a graphical representation showing color change results for an article having been exposed to high temperature, humidity and direct UV light with and without the container of FIG. **1**; and
[0020] FIG. **11** is a process flow showing a method of protecting an article using the container of FIG. **1**.
[0021] Corresponding reference characters indicate corresponding elements among the view of the drawings. The headings used in the figures do not limit the scope of the claims.
DETAILED DESCRIPTION
[0022] The present technology may be described in terms of functional block components and various processing steps. Such functional blocks may be realized by any number of components configured to perform the specified functions and achieve the various results. For example, the present technology may employ various types of materials, gases, valves, and the like, which may carry out a variety of functions. In addition, the present technology may be practiced in conjunction with any number of applications, and the apparatus described is merely one exemplary application for the technology.
[0023] Various representative implementations of the present technology may be used to store, preserve, and protect footwear, for example, sneakers or sports shoes. In addition, embodiments

may be applicable for housing and storing any other collectable items that contain oxygen and/or moisture-sensitive materials, such as comic books, figurines, dolls, sports cards, game playing cards, books, documents, photographs, and the like. Embodiments of the present technology may protect and preserve items by preventing decay and deterioration of the item due to light, temperature, dirt, mold, humidity, hydrolysis due to moisture in the air, and oxidation with exposure to air.

[0024] Referring now to FIG. **1**, in an exemplary embodiment of the present technology, an apparatus as described herein may take the form of at least one container **100** that defines an interior cavity **102** in communication with an opening **146** formed by a container body **104**. In general, the container **100** receives at least one article **10** within the interior cavity **102** for preservation of the article **10**. In one embodiment, the container **100** includes an access door **106** that provides access to the interior cavity **102** along the opening **146** of the container **100**. The access door **106** includes a sealing mechanism **160** that couples the access door **106** along the opening **146** such that the container **100** is impermeable to ambient air (i.e., airtight). The sealing mechanism **160** can include, without limitation, a zipper arrangement, or other such airtight sealing component. The container **100** includes an atmospheric control mechanism **110** to maintain an internal atmospheric composition that preserves the article **10** or otherwise prevents degradation of the article **10** due to various factors. In some embodiments, the container body **104** can include a plurality of container walls (or container panels) arranged to form the interior cavity **102** of the container **100**.

[0025] In a first example shown in FIGS. **2**A-**5**B, the container **100** can take the form of a pillow-like configuration constructed using two container walls. In another example shown in FIGS. **6**A-**9**B, a container **200** may have a cube or otherwise box-like shape configuration with six container walls. Further, in some embodiments, the container **100** (and/or **200**) can include an outer container **180** that receives and at least partially encapsulates the container **100**.

[0026] With continued reference to FIG. **1**, the atmospheric control mechanism **110** of the container **100** includes an inlet valve **112** coupled to the interior cavity **102** that enables one-way fluid flow of a selected gas into the interior cavity **102**. The selected gas can be applied to maintain the internal atmospheric composition of the interior cavity **102** such that the internal atmospheric composition of the interior cavity **102** of the container **100** is free from various contaminants including dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected, as further described herein. The inlet valve **112** can be a one-way valve that engages with an external canister **120** that provides the selected gas to accommodate one-way fluid flow communication of the selected gas from the external canister **120** into the interior cavity **102**. As shown, the inlet valve **112** can engage with the external canister **120** through an inflator device **122** that propels the selected gas from the external canister **120** into the interior cavity **102** or otherwise establishes an “airtight” connection between the external canister **120** and the interior cavity **102**. Further, the atmospheric control mechanism **110** of the container **100** includes an outlet valve **114** coupled to the interior cavity **102** that establishes one-way fluid flow communication from the interior cavity **102** to the external environment to purge the interior cavity **102** of air that includes contaminants such as dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected. In an operation which will be described in greater detail below, the outlet valve **114** can be configured for manual or automatic release (e.g., can “open” to release air when an air pressure within the interior cavity **102** meets and/or exceeds a predetermined threshold value). Collectively, the atmospheric control mechanism **110** of the container **100** maintains the internal atmospheric composition of the interior cavity **102** such that the internal atmospheric composition of the interior cavity **102** of the container **100** is free from various contaminants including dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected.

[0027] With reference to FIGS. **2**A-**3**C, the container body **104** of the container **100** includes at least a front panel **142** and a rear panel **144** that collectively encapsulate the interior cavity **102**. In the example shown, the front panel **142** and the rear panel **144** can be of a flexible material to enable the container **100** to accommodate articles of varying shapes and sizes. As illustrated, the front panel **142** can include the atmospheric control mechanism **110** including the inlet valve **112** and the outlet valve **114**. The front panel **142** can also provide the opening **146** of the interior cavity **102** including the access door **106** that provides access to the interior cavity **102** along the opening **146** of the container **100**. The access door **106** includes the sealing mechanism **160** that couples the access door **106** along the opening **146** to prevent permeation of ambient air (i.e., airtight) such that air cannot enter or leave the interior cavity **102** through the access door **106** when the access door **106** is sealed. In the example shown, the sealing mechanism **160** includes an airtight zipper **162**, although various other embodiments of the sealing mechanism **160** are contemplated including a gasket, a press-seal mechanism, and/or an adhesive mechanism. As illustrated, the rear panel **144** can include a transparent portion **164** that enables viewing of the article **10** encapsulated within the interior cavity **102**. Further, the container body **104** can include a handle **148** for ease of transport and handling of the container **100**. As shown in FIGS. **3**A-**3**C, the container **100** can inflate upon introduction of the selected gas into the interior cavity **102**. As shown, when inflated, the container **100** can provide cushioning to prevent the article **10** within the interior cavity **102** from being crushed.

[0028] In addition, the container body **104** of the container **100** can be formed using a material that is impenetrable by air, such as plastic, metal, and the like.

[0029] In one embodiment, the container body **104** can be formed using a flexible material that provides a moveable structure and shape. For example, the container body **104** can be formed using flexible PVC, rubber, TPU, silicon, and the like, and can further be lined with one or more insulating and/or sealing materials such as BoPET (biaxially-oriented polyethylene terephthalate) (e.g., Mylar). The container body **104** can also be formed using a woven fiber material such as cotton, polyester and/or nylon that can be coated with a sealing material to provide airtight resistance to chemicals, temperature, ozone, ultraviolet (UV) radiation, mold, mildew, and abrasion.

[0030] In some embodiments, a material used to form the container body **104** can be opaque, in particular the front panel **142** and in some embodiments a portion of the rear panel **144**. In other embodiments, a material used to form the transparent portion **164** of the container body **104** can be transparent or translucent and can include an UV blocking property. For example, the UV blocking property can be integrated within the material or can include a film having a UV blocking property that can adhere to the container body **104**.

[0031] The container **100** can further include the outer container **180** that forms an outer layer **182** configured to surround the container **100**. In other words, the container **100** can be disposed within (e.g., nested within) the outer container **180**. The outer layer **182** can have a similar overall shape as the container **100** but may have larger dimensions than the container **100** to allow the container **100** to fit within the outer layer **182**. In some embodiments, the outer layer **182** can be of a rigid material, or can be of a flexible material. In the embodiment of the container **100** being of a flexible material, it can be advantageous for the outer layer **182** to be of a rigid material to prevent crushing of the container **100** and/or article **10** enclosed. In some embodiments, the outer layer **182** can include at least one removable panel (not visible, see removable panel **284** of FIGS. **6**C and **6**D) that can be completely or partially removable from the remaining portions of the outer layer **182**. For example, the removable panel may connect to other portions of the outer layer **182** using any suitable fastener, such as Velcro, pins, buttons, snaps, slide fasteners, buckles, zippers, hooks and eyes, ties, or any combination thereof. The outer layer **182** can include any suitable material that provides light/UV shielding, abrasion resistance, and/or desirable aesthetics. For example, the outer layer **182** may comprise a nylon material, polyester material, cotton material, or the like.

[0032] The outer layer **182** can further include a first aperture **186** adapted to accommodate the inlet valve **112** and a second aperture **188** adapted to accommodate the outlet valve **114**. For example, when the container **100** is placed inside the outer layer **182**, the inlet valve **112** can extend through the first aperture **186** and the outlet valve **114** can extend through the second aperture **188**.

[0033] With reference to FIGS. **4**A and **4**B, the atmospheric control mechanism **110** of the container **100** includes the inlet valve **112** coupled to the interior cavity **102** that enables one-way fluid flow of the selected gas into the interior cavity **102**. The inlet valve **112** can be a one-way valve that couples with the external canister **120** that provides the selected gas to accommodate one-way fluid flow communication of the selected gas from the external canister **120** into the interior cavity **102**. For example, the inlet valve **112** can be a one-way valve such as a Schrader valve, a Presta valve and/or a Dunlop valve. In some embodiments, the external canister **120** can indirectly couple with the inlet valve **112** by the inflator device **122** that propels the contents of the external canister **120** into the interior cavity **102** through the inlet valve or otherwise establishes an "airtight" connection between the external canister **120** and the inlet valve **112**. The inlet valve **112** can be configured to allow the selected gas to enter the interior cavity **102** when the inlet valve **112** is directly or indirectly coupled with the external canister **120**, and can further be configured to prevent ambient air from an external environment from entering the interior cavity **102** and to prevent expulsion of gas from the interior cavity **102** out of the inlet valve **112**.

[0034] The selected gas can be selected to maintain the internal atmospheric composition such that the internal atmospheric composition of the interior cavity **102** of the container **100** is free from various contaminants including dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected. In a primary embodiment, the selected gas includes at least one inert gas (e.g., a non-reactive gas). For example, the selected gas can include noble gases such as argon and/or other inert gases such as purified nitrogen. The selected gas is devoid of moisture or oils to prevent moisture and/or oil-related degradation or contamination of the article **10**. In some embodiments, an amount of the selected gas within the external canister **120** can be sized such that, when released, the external canister **120** provides an equal or greater volume of the selected gas than a volumetric capacity of the interior cavity **102** of the container **100**. In one example, the external canister **120** can include 6.5 g of the selected gas, which can be more than enough to fill the entire interior cavity **102** if the inflated dimensions of the container **100** are 343 mm by 251 mm by 130 mm. As such, the external canister **120** can flush or purge the interior cavity **102** and the majority of the internal atmospheric composition within the interior cavity **102** will be left with the selected gas.

[0035] Further, with reference to FIGS. **5**A-**5**E, the atmospheric control mechanism **110** of the container **100** includes the outlet valve **114** coupled to the interior cavity **102** that establishes one-way fluid flow communication from the interior cavity **102** to the external environment to purge the interior cavity **102** of air that includes contaminants such as dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected. As such, the outlet valve **114** further contributes to maintaining the internal atmospheric composition. In particular, the outlet valve **114** can assume a first "closed" configuration (FIG. **5**B) that prevents intake of air from the external environment into the interior cavity **102** and prevents expulsion of pressurized air from the interior cavity **102** when an air pressure within the interior cavity **102** is below a predetermined threshold value. Further, the outlet valve **114** can be configured to release pressurized air within the interior cavity **102** and in some embodiments can include a relief mechanism **116** (FIG. **5**E) that opens the outlet valve **114** to a second "venting" configuration (FIG. **5**D) when an air pressure within the interior cavity **102** meets or exceeds the predetermined threshold value. The outlet valve **114** can also be configured for manual operation such that a user can "toggle" the outlet valve **114** between the first "closed" configuration and the second "venting" configuration as needed to purge the interior cavity **102** of air that includes contaminants. In some embodiments, the outlet valve **114** can be configured for modulation of a flow rate of air being expelled through the outlet valve **114** by constricting or dilating a venting path through the outlet valve **114** until a desired flow rate is reached. For example, an intermediate "constricted venting" configuration is shown in FIG. **5**C where an indicator **118** of the outlet valve **114** shows an intermediate position between the "closed" configuration of FIG. **5**B and the "venting" configuration of FIG. **5**D, resulting in an intermediate flow rate. Further, in some embodiments, the relief mechanism **116** of the outlet valve **114** can be configured for manual adjustment such that a user can set and/or modulate the predetermined threshold value at which the relief mechanism **116** opens and/or closes the outlet valve **114**.

[0036] During operation of the atmospheric control mechanism **110**, the selected gas enters the interior cavity **102** of the container **100** through the inlet valve **112**, which increases an air pressure within the interior cavity **102**. The outlet valve **114** can be in the first "closed" configuration until the air pressure reaches a predetermined threshold, at which the relief mechanism **116** of the outlet valve **114** activates and the outlet valve **114** assumes the second "venting" configuration. In another aspect, the outlet valve **114** can be manually transitioned between the first "closed" configuration and the second "venting" configuration prior to or during introduction of the selected gas into the interior cavity **102**. Pressurized air, including air that includes contaminants mentioned above, exits the interior cavity **102** through the outlet valve **114** when the outlet valve is in the second "venting" configuration. When the outlet valve **114** is in the second "venting" configuration, the outlet valve **114** purges air that includes contaminants from the interior cavity **102** as the interior cavity **102** is filled with the selected gas (e.g., the force of the selected gas entering the interior cavity **102** at the inlet valve **112** displaces the air within the interior cavity **102** and forces the air out of the outlet valve **114**).

[0037] The dilution and venting of the oxygen and/or humidity components in the captured air reduces the oxygen and moisture levels within the interior cavity and continues to do so as long as the selected gas is being forced into the interior cavity **102**. Once the external canister **120** is separated from the inlet valve **112**, gas dispensing stops and the inlet valve **112** seals itself to provide an internal atmospheric composition that is both dry and inert inside the interior cavity **102**. Once the excess internal pressure of the interior cavity **102** is equalized through venting at the outlet valve **114**, the outlet valve **114** seals itself to prevent external air from entering the interior cavity **102**. The result of this air and moisture purging, the airtight nature of the container **100**, and the opaque nature of the outer container **180** is that the contents of the container **100** are mostly or completely free of oxygen and water and are also shielded from visible and UV light. In some embodiments, the container **100** can include desiccant materials or modules thereof, and can in some embodiments include pockets (not visible, see pocket **299** of FIG. **9**B) to hold such modules as an added method of protection from moisture. Further, the container **100** can include modules of oxygen scavenging materials able to consume any remaining oxygen in the container **100** after the initial purging.

[0038] With reference to FIGS. **6**A-**8**C, a container **200** can include aspects of the container **100**, however a container body **204** of the container **200** can be of a box-like configuration that can include a rigid material such as metal or plastic. Similar to that of the container **100**, the container **200** defines an interior cavity **202** in communication with an opening **246** formed by the container body **204** and can receive the article **10** therein for preservation of the article **10**. In a primary embodiment, the container **200** includes an access door **206** that provides access to the interior cavity **202** along the opening **246** of the container **200**. The access door **206** includes a sealing mechanism **260** that couples the access door **206** along the opening **246** to prevent permeation of ambient air into the interior cavity **202** of the container **200** (i.e., airtight). In another aspect, the container **200** can include an atmospheric control mechanism **210** to maintain an internal atmospheric composition that preserves the article **10** or otherwise prevents degradation of the article **10** due to various factors. The container body **204** can include a plurality of container walls (or container panels) arranged to form the interior cavity **202** of the container **200**. Further, in some embodiments, the container **200** can include an outer container **280** that receives and at least partially encapsulates the container **200**; in some embodiments the outer container **280** can be of an opaque material to provide additional shielding against UV radiation.

[0039] With additional reference to FIGS. **9**A and **9**B, the atmospheric control mechanism **210** of the container **200** includes an inlet valve **212** coupled to the interior cavity **202** that enables one-way fluid flow of a selected gas into the interior cavity **202**. The selected gas can be selected to maintain the internal atmospheric composition of the interior cavity **202** such that the internal atmospheric composition of the interior cavity **202** of the container **200** is free from various contaminants including dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected. Similar to that of the inlet valve **112** described above with respect to FIGS. **4**A and **4**B, the inlet valve **212** can be a one-way valve that engages with an external canister **220** that provides the selected gas to accommodate one-way fluid flow communication of the selected gas from the external canister **220** into the interior cavity **202**. As shown, the inlet valve **212** can couple with the external canister **220** through an inflator device **222** that propels the selected gas from the external canister **220** into the interior cavity **202** or otherwise establishes an “airtight” connection between the external canister **220** and the interior cavity **202**. Further, the atmospheric control mechanism **210** of the container **200** includes an outlet valve **214** coupled to the interior cavity **202** that establishes one-way fluid flow communication from the interior cavity **202** to the external environment to purge the interior cavity **202** of air that includes contaminants such as dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected. Similar to that of the outlet valve **114** described above with respect to FIGS. **5**A-**5**E, the outlet valve **214** can be configured for manual or automatic release (e.g., can “open” to release air when an air pressure within the interior cavity **202** meets and/or exceeds a predetermined threshold value). Collectively, the atmospheric control mechanism **210** of the container **200** maintains the internal atmospheric composition of the interior cavity **202** such that the internal atmospheric composition of the interior cavity **202** of the container **200** is free from various contaminants including dust, oxygen, water, atmospheric moisture (humidity), reactive oxygen species, airborne chemical pollutants and/or airborne chemicals not conducive to the longevity or preservation of the article **10** to be protected.

[0040] With reference to FIGS. **8**A-**8**C, the container body **204** of the container **200** includes at least a top panel **242**, a plurality of side panels **248** and a bottom panel **244** that collectively encapsulate the interior cavity **202**. In the example shown, the top panel **242** forms the access door **206** and is configured for partial or complete removal from the remainder of the container body **204** to provide access to the interior cavity **202** therein. In the example shown, while the plurality of side panels **248** and a bottom panel **244** can be formed from a rigid material, the top panel **242** can include a flexible material that forms the access door **206** that provides access to the interior cavity **202** along the opening **246** of the container **200**. Alternatively, the top panel **242** can include a rigid material. The access door **206** includes the sealing mechanism **260** that couples the access door **206** along the opening **246** to ensure that the interior cavity **202** of the container **200** is airtight when the access door **206** is closed. For example, the sealing mechanism **260** of the container **200** and/or the access door **206** can include an airtight zipper **262**, although various other embodiments of the sealing mechanism **260** are contemplated including a press-seal mechanism, an adhesive mechanism, rubber fittings or gaskets to prevent ambient air from the external environment from entering the container **200**. As illustrated, the top panel **242** can include a first transparent portion

264 that enables viewing of the article **10** encapsulated within the interior cavity **202**. Further, in some embodiments, at least one of the plurality of side panels **248** can include a second transparent portion **249** that enables viewing of the article **10** encapsulated within the interior cavity **202**. As illustrated, the plurality of side panels **248** can include the atmospheric control mechanism **210** including the inlet valve **212** and the outlet valve **214**. The plurality of side panels **248** can also collectively form the opening **246** of the interior cavity **202** that couples with the access door **206** formed by the top panel **242**.

[0041] In addition, the container body **204** of the container **200** can be formed using a material that is impenetrable by air, such as plastic, metal, and the like. In one embodiment, the container body **204** can be formed using a flexible material that provides a moveable structure and shape. For example, the container body **204** can be formed using flexible PVC, rubber, TPU, silicon, and the like, and can further be lined with one or more insulating and/or sealing materials such as BoPET (biaxially-oriented polyethylene terephthalate) (e.g., Mylar). The container body **204** can also be formed using a woven fiber material such as cotton, polyester and/or nylon that can be coated with a sealing material to provide airtight resistance to chemicals, temperature, ozone, ultraviolet (UV) radiation, mold, mildew, and abrasion.

[0042] In some embodiments, a material used to form the side panels **248** and the bottom panel **244** of the container body **204** can be opaque. Further, a material used to form the first transparent portion **264** and the second transparent portion **249** of the container body **204** (e.g, of the top panel **242**) can be transparent or translucent and can include a UV blocking property. For example, the UV blocking property can be integrated within the material or can include a film having a UV blocking property that can adhere to the container body **204**.

[0043] In operation, similar to that of the atmospheric control mechanism **110** described above with reference to FIGS. **4**A-**5**D, during operation of the atmospheric control mechanism **210**, the selected gas enters the interior cavity **202** of the container **200** through the inlet valve **212**, which increases an air pressure within the interior cavity **202**. The outlet valve **214** can be in a first “closed” configuration until the air pressure reaches a predetermined threshold, at which a relief mechanism of the outlet valve **214** activates and the outlet valve **214** assumes the second “venting” configuration. In another aspect, the outlet valve **214** can be manually transitioned between the first “closed” configuration and the second “venting” configuration prior to or during introduction of the selected gas into the interior cavity **202**. Pressurized air, including air that includes contaminants mentioned above, exits the interior cavity **202** through the outlet valve **214** when the outlet valve **214** is in the second “venting” configuration. When the outlet valve **214** is in the second “venting” configuration, the outlet valve **214** purges air that includes contaminants from the interior cavity **202** as the interior cavity **202** is filled with the selected gas (e.g., the force of the selected gas entering the interior cavity **202** at the inlet valve **212** displaces the air within the interior cavity **202** and forces the air out of the outlet valve **214**).

[0044] The dilution and venting of the oxygen and/or humidity components in the captured air reduces the oxygen and moisture levels within the interior cavity **202** and continues to do so as long as the inlet gas is being forced in from the external canister **220** through the inlet valve **212**, with an intermediate connection therebetween optionally provided by the inflator device **222**. Once the external canister **220** is separated from the inlet valve **212**, gas dispensing stops and the inlet valve **212** seals itself to provide an internal atmospheric composition that is both dry and inert inside the interior cavity **202**. Once the excess internal pressure of the interior cavity **202** is equalized through venting at the outlet valve **214**, the outlet valve **214** seals itself to prevent external air from entering the interior cavity **202**. The result of this air and moisture purging provided by the atmospheric control mechanism **210**, in addition to the airtight nature of the container **200**, and the opaque nature of the outer container **280** is that the contents of the container **200** are mostly or completely free of contaminants such as oxygen and water and are also shielded from visible and UV light. In some embodiments, the container **200** can include desiccant materials or modules thereof as an added method of protection from moisture. Further, the container **200** can include modules of oxygen scavenging materials able to consume any remaining oxygen in the apparatus after the initial purging.

[0045] With reference to FIGS. **6**A-**7**, the container **200** can further include the outer container **280** that forms an outer layer **282** configured to surround the container **200**. In other words, the container **200** can be disposed within (e.g., nested within) the outer container **280** as shown in the sequence of FIGS. **6**A-**6**D and in the exploded view of FIG. **7**. The outer layer **282** can have a similar overall shape as the container **200** but may have larger dimensions than the container **200** to allow the container **200** to fit within the outer layer **282**. For example, in the case of the container **200** having a box-shaped configuration having four side panels **248**, the top panel **242** and the bottom panel **244**, the outer layer **282** can include six panels including four outer side panels **292**, an outer top panel **294** and an outer bottom panel **296** arranged to receive the container **200**. In some embodiments, the outer layer **282** can be of a rigid material such as metal or plastic, but can alternatively be of a flexible material such as woven fiber, flexible plastic, and/or a combination thereof. In some embodiments, the outer layer **282** can include at least one removable panel **284** that can be completely or partially removable from the remaining portions of the outer layer **282**, as shown in the sequence of FIGS. **6**A-**6**D. For example, the removable panel **284** may connect to other portions of the outer layer **282** using any suitable fastener, such as Velcro, pins, buttons, snaps, slide fasteners, buckles, zippers, hooks and eyes, ties, or any combination thereof. The outer layer **282** can include any suitable material that provides light/UV shielding, abrasion resistance, and/or desirable aesthetics. For example, the outer layer **282** may comprise a nylon material, polyester material, cotton material, or the like.

[0046] The outer layer **282** can further include a first aperture (not visible, see first aperture **186** of FIG. **1**) adapted to accommodate the inlet valve **212** and a second aperture **288** adapted to accommodate the outlet valve **214**. For example, when the container **200** is placed inside the outer layer **282**, the inlet valve **212** can extend through the first aperture and the outlet valve **214** can extend through the second aperture **288**. Further, the outer layer **282** can include an outer handle **298** for ease of transport and handling of the container **200**.

[0047] In various embodiments, as shown in the exploded view of FIG. **7** and as illustrated in FIG. **9**B, the container **200** (and/or the container **100** of FIGS. **1**-**3**C) can include a pocket **299** along the interior cavity **202** of the container **200**, on an exterior surface of the container **200**, and/or on an exterior surface of the outer layer **282**. For example, the pocket **299** can be attached to any wall of the container **200** or the outer layer **282**. In various embodiments, the pocket **299** can include a mesh material or any other material suitable for securing various items, such as shoelaces, documents, and the like. In some embodiments, the pocket **299** can include desiccant materials or modules thereof, and can further include modules of oxygen scavenging materials able to consume any remaining oxygen in the interior cavity **202** after the initial purging.

[0048] With reference to FIG. **10**, the ability of the container **100** to protect articles from environmental changes was examined via accelerated environmental testing in a QSUN chamber. Xenon arc exposure was employed per ASTM G155-21, cycle 12 for 400 hours. Experimentation results are provided with respect to a pair of Nike Air Force 1 shoes that were exposed to four separate 100 hour cycles of high temperature, humidity and direct UV light, with one being unshielded and another being stored within the container **100** during exposure. A color configuration of each respective shoe was examined with respect to their original color configuration across four 100 hour intervals. As visible within the plot of FIG. **10**, color degradation for the exposed shoe was significantly higher than that of the shielded shoe that was stored within the container **100**. Color evaluations were performed with a spectrophotometer per ASTM D2244-21 every 100 hours until completion. Cycle 12 of ASTM G155-21 employs 0.35 W/(m2 nm) @ 340 nm, cycling through 18 hours of light to 6 hours of dark with the chamber air temperature and relative humidity at 30% RH 47° C. and 90% RH and 35° C., respectively for the

light and dark cycles.
[0049] As such, the container **100** and/or **200** can create an environment that is free of oxygen and water and is shielded from visible and ultraviolet light. In addition, the container **100** and/or **200** prevents degradation of articles stored therein by preventing oxidation (chemical breakdown of the material via a reaction with the oxygen in the air), hydrolysis (chemical breakdown of the polymer structures via a reaction with water in the air), photo-degradation (breakdown of the polymer structures due to ultraviolet electromagnetic radiation), and photo-oxidation (breakdown of the polymer chemical structures due to light-activated oxidation of the chemical bonds). Furthermore, the container **100** and/or **200** is able to maintain a desired internal atmospheric composition and hold the pressure during a purging process. In some embodiments, the article **10** is a shoe such as a Nike Air Force 1 shoe. The internal atmospheric composition associated with the selected gas reduces a change in the color configuration of the article **10** over time. The article **10** can include a sole made of plastic or rubber material (e.g., a polyurethane material that covers a midsole of the shoe) that can degrade over time, and the container **100** or **200** encloses the shoe within the interior cavity and protects the shoe from the external environment. In particular, the internal atmospheric composition associated with the selected gas reduces a change in the integrity of the sole of the shoe over time. Further, the article **10** can include a shoe that includes an adhesive that couples various components of the shoe that can also degrade over time, and the container **100** or **200** encloses the shoe within the interior cavity and protects the shoe from the external environment. The internal atmospheric composition associated with the selected gas reduces a change in the adhesive properties of the adhesive of the shoe over time. Note that while various embodiments herein are described with respect to a shoe being the article **10**, the article **10** can include other items of value that can be protected by the container **100** and/or **200** can include but are not limited to clothing, official documents such as government-issued certificates (e.g., birth certificates, social security cards, passports) and certificates of authenticity, jewelry, watches, electronic items (e.g., game cartridges, compact discs, tapes, gaming systems, computing components, cameras), cards (e.g., Pokémon cards, baseball cards), books, figurines, photography and film materials (e.g., developed and/or undeveloped photographic film).
[0050] FIG. **11** shows a method **300** of protecting an article using the container **100** and/or **200**. Block **310** of the method **300** includes opening the access door of the container. Block **320** includes placing the article inside the interior cavity of the container. Block **330** includes sealing the access door of the container by the sealing mechanism. Block **340** includes opening the outlet valve to the venting configuration. Block **350** includes coupling the external canister with the inlet valve. Block **360** includes filling the interior cavity of the container with the selected gas of the external canister. Note that block **340** can be performed automatically after block **360** by the relief mechanism of the outlet valve. Block **370** includes purging excess air and contaminants from the interior cavity of the container through the outlet valve. Note that block **370** is resultant of introducing the selected gas into the internal container at block **360**. Block **380** includes closing the outlet valve to the closed configuration and/or closing the inlet valve.
[0051] The particular implementations shown and described are illustrative of the technology and its best mode and are not intended to otherwise limit the scope of the present technology in any way. Indeed, for the sake of brevity, conventional manufacturing, connection, preparation, and other functional aspects of the apparatus may not be described in detail. Furthermore, the connecters and points of contact shown in the various figures are intended to represent exemplary physical relationships between the various elements. Many alternative or additional functional relationships or physical connections may be present in a practical system.
[0052] In the foregoing description, the technology has been described with reference to specific exemplary embodiments. Various modifications and changes may be made, however, without departing from the scope of the present technology as set forth. The description and figures are to be regarded in an illustrative manner, rather than a restrictive one and all such modifications are

intended to be included within the scope of the present technology. Accordingly, the scope of the technology should be determined by the generic embodiments described and their legal equivalents rather than by merely the specific examples described above. For example, the steps recited in any method or process embodiment may be executed in any appropriate order and are not limited to the explicit order presented in the specific examples. Additionally, the components and/or elements recited in any system embodiment may be combined in a variety of permutations to produce substantially the same result as the present technology and are accordingly not limited to the specific configuration recited in the specific examples.
[0053] Benefits, other advantages and solutions to problems have been described above with regard to particular embodiments. Any benefit, advantage, solution to problems or any element that may cause any particular benefit, advantage or solution to occur or to become more pronounced, however, is not to be construed as a critical, required or essential feature or component.
[0054] The present technology has been described above with reference to an exemplary embodiment. However, changes and modifications may be made to the exemplary embodiment without departing from the scope of the present technology. These and other changes or modifications are intended to be included within the scope of the present technology.

Claims

1. An apparatus for preservation of footwear, comprising: at least one container configured to maintain an internal atmospheric composition to preserve an article of footwear, the at least one container including: a body comprising an interior cavity in communication with an opening configured to receive the article of footwear, the body defining at least a first panel and a second panel that collectively encapsulate the interior cavity, both of the first panel and the second panel of the body being formed of a multi-layered flexible material, wherein the flexible material of the body includes a woven fiber coated with a sealing material that provides a moveable structure and shape to accommodate varying shapes and sizes for the article of footwear while also providing airtight resistance to chemicals, temperature, ozone, ultraviolet (UV) radiation, mold, mildew, and abrasion for the article of footwear; an access door that provides access to the interior cavity along the opening of the at least one container, the access door including a sealing mechanism that couples the access door along the opening such that the sealing mechanism prevents permeation of air from an external environment into the interior cavity; an inlet valve coupled to the interior cavity configured to provide one-way fluid flow of a selected gas from an inflator device into the interior cavity to reach and maintain the internal atmospheric composition and preserve the article of footwear; and an outlet valve coupled to the interior cavity that establishes one-way fluid flow communication from the interior cavity to the external environment, the outlet valve configured to modulate a predetermined flow of air being expelled through the outlet valve.
2. The apparatus of claim 1, wherein the first panel comprises an opaque material, and wherein at least some of the second panel comprises a transparent material that defines a transparent portion, the transparent portion including an ultraviolet (UV) blocking property, the transparent portion defined opposite the first panel to accommodate viewing of the article along a discrete portion of the second panel.
3. The apparatus of claim 1, wherein the outlet valve is configured for manual control to modulate the flow of rate of air being expelled through the outlet valve by constricting or dilating a venting path through the outlet valve until a desired flow rate is reached.
4. The apparatus of claim 1, wherein the selected gas includes at least one noble or inert gas.
5. The apparatus of claim 1, wherein the selected gas is devoid of moisture or oils.
6. The apparatus of claim 1, further comprising an outer container, the at least one container being disposed within and at least partially encapsulated by the outer container.
7. The apparatus of claim 1, wherein the inlet valve is a one-way pressure release valve.

8. The apparatus of claim 1, wherein the inlet valve engages with an external canister that provides the selected gas to accommodate one-way fluid flow communication of the selected gas from the external canister into the interior cavity of the at least one container.
9. The apparatus of claim 1, wherein the outlet valve is configured for manual operation.
10. The apparatus of claim 1, wherein the outlet valve is configured to release pressurized air within the interior cavity when an air pressure within the interior cavity reaches a predetermined threshold value.
11. The apparatus of claim 1, wherein the at least one container is configured to limit or prevent ultraviolet radiation from entering the interior cavity.
12. The apparatus of claim 1, further comprising an outer layer that encloses the at least one container therein.
13. The apparatus of claim 12, wherein the outer layer includes at least one removable panel configured for partial or complete removal from the outer layer.
14. The apparatus of claim 1, wherein the article of footwear includes a shoe including a color configuration, such that the at least one container encloses the shoe within the interior cavity and protects the shoe from the external environment.
15. The apparatus of claim 14, wherein the internal atmospheric composition associated with the selected gas reduces a change in the color configuration of the shoe over time.
16. The apparatus of claim 1, wherein the article of footwear includes a shoe including a sole including a plastic or rubber material, such that the at least one container encloses the shoe within the interior cavity and protects the shoe from the external environment.
17. The apparatus of claim 16, wherein the internal atmospheric composition associated with the selected gas reduces a change in integrity of the sole over time.
18. The apparatus of claim 1, wherein the article of footwear includes a shoe having an adhesive, such that the at least one container encloses the shoe within the interior cavity and protects the shoe from the external environment.
19. The apparatus of claim 18, wherein the internal atmospheric composition associated with the selected gas reduces a change in the adhesive of the shoe over time.
20. A method of making an apparatus for preserving an article, comprising: forming at least one container configured to maintain an internal atmospheric composition, the at least one container including an interior cavity in communication with an opening configured to receive an article; forming an access door along the opening of the at least one container that provides air-tight access to the interior cavity; coupling an inlet valve to the interior cavity that provides one-way fluid flow of a selected gas into the interior cavity; and coupling an outlet valve to the interior cavity that provides one-way fluid flow communication from the interior cavity to an external environment.
21. The method of claim 20, further comprising: connecting an inflator device including an external canister to the inlet valve to provide the selected gas into the interior cavity.
22. An apparatus for preservation of at least one article, comprising: at least one container comprising: a body including an interior cavity in communication with an opening configured to receive an article; an access door that provides access to the interior cavity along the opening of the at least one container; an inlet valve positioned along the body that provides one-way fluid flow of a one or more gases into the interior cavity; and a manually adjustable outlet valve positioned along the body that provides one-way fluid flow communication from the interior cavity to an external environment, wherein the body comprises a flexible material that provides a moveable structure and shape.
23. The apparatus of claim 22, wherein the flexible material of the body includes a sealing material to provide airtight resistance to the external environment as well as long term resistance to air and moisture permeation via molecular penetration through composite layers of the body.
24. The apparatus of claim 22, wherein the flexible material of the body includes material that provides UV protection from the external environment.

25. The apparatus of claim 22, wherein the outlet valve is configured for manual control to modulate a flow of rate of air being expelled through the outlet valve such that the outlet valve accommodates constricting or dilating a venting path through the outlet valve until a desired flow rate is reached.